

Daily Integral - Medium

$$\int_0^1 \tan^{-1} \left(\frac{2x-1}{1+x-x^2} \right) dx$$

This problem was generously submitted by Ved Khatri on <https://dailyintegral.com>.

Solution

First, we take $2x - 1 = x - (1 - x)$ and $1 + x - x^2 = 1 + x(1 - x)$:

$$\int_0^1 \tan^{-1} \left(\frac{x - (1 - x)}{1 + x(1 - x)} \right) dx.$$

Now, we can use the property for the difference of inverse tangents, $\tan^{-1}(a) - \tan^{-1}(b) = \tan^{-1}((a-b)/(1+ab))$, with $a = x$ and $b = 1 - x$:

$$\begin{aligned} & \int_0^1 (\tan^{-1}(x) - \tan^{-1}(1-x)) dx \\ &= \int_0^1 \tan^{-1}(x) dx - \int_0^1 \tan^{-1}(1-x) dx. \end{aligned}$$

Now, we can use King's rule which states that $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$, on the first integral:

$$\begin{aligned} & \int_0^1 \tan^{-1}(1-x) dx - \int_0^1 \tan^{-1}(1-x) dx \\ &= 0. \end{aligned}$$